PEN-240840116006

Practical-2 Perform Data inspection, cleaning,banding missing values,dupliate removal and data transforation using pandas.

Step: 1 Import Pandas

```
import pandas as pd
```

Step: 2 Create Dataframe

```
data ={
    "Id":[101,102,103,104,104,105,106,107],
    "Name":["Rahul","Priya","Amit",None,"Riya","kavya","darshana","himay"],
    "Age":[20,21,None,20,20,22,23,28],
    "City":["Surat","Navsari","Surat","Bharuch","Bharuch","Navsari","Surat","Ahemedabad"],
    "Marks":[85,None,78,88,88,81,90,93]
}
```

```
df = pd.DataFrame(data)
df
```

	Id	Name	Age	City	Marks
0	101	Rahul	20.0	Surat	85.0
1	102	Priya	21.0	Navsari	NaN
2	103	Amit	NaN	Surat	78.0
3	104	None	20.0	Bharuch	88.0
4	104	Riya	20.0	Bharuch	88.0
5	105	kavya	22.0	Navsari	81.0
6	106	darshana	23.0	Surat	90.0
7	107	himay	28.0	Ahemedabad	93.0

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Step: 3 Inspect the Database

```
# Display first 5 row of dataframe.
df.head()
```

	Id	Name	Age	City	Marks
0	101	Rahul	20.0	Surat	85.0
1	102	Priya	21.0	Navsari	NaN
2	103	Amit	NaN	Surat	78.0
3	104	None	20.0	Bharuch	88.0
4	104	Riya	20.0	Bharuch	88.0

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```
# Display last 5 row of dataframe
df.tail()
```

	Id	Name	Age	City	Marks
3	104	None	20.0	Bharuch	88.0
4	104	Riya	20.0	Bharuch	88.0
5	105	kavya	22.0	Navsari	81.0
6	106	darshana	23.0	Surat	90.0
7	107	himay	28.0	Ahemedabad	93.0

```
# Display detail info of dataframe
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8 entries, 0 to 7
Data columns (total 5 columns):
 #   Column  Non-Null Count  Dtype  
---  ------  --------------  -----  
 0   Id      8 non-null      int64  
 1   Name    7 non-null      object 
 2   Age     7 non-null      float64
 3   City    8 non-null      object 
 4   Marks   7 non-null      float64
dtypes: float64(2), int64(1), object(2)
memory usage: 452.0+ bytes
```

```
# Generate statistical info.
df.describe()
```

	Id	Age	Marks
count	8.00	7.000000	7.000000
mean	104.00	22.000000	86.142857
std	2.00	2.886751	5.209881
min	101.00	20.000000	78.000000
25%	102.75	20.000000	83.000000
50%	104.00	21.000000	88.000000
75%	105.25	22.500000	89.000000
max	107.00	28.000000	93.000000

```
# Display dimension of dataframe.
df.shape
```

```
(8, 5)
```

```
# Display name of column.
df.columns
```

```
Index(['Id', 'Name', 'Age', 'City', 'Marks'], dtype='object')
```

```
# Display datatype of each column.
df.dtypes
```

	0
Id	int64
Name	object
Age	float64
City	object
Marks	float64

dtype: object

Step 4 : Check missing Values.

```
# Check Missing (null) values in every column.
df.isnull().sum()
```

	0
Id	0
Name	1
Age	1
City	0
Marks	1

dtype: int64

Step 5: Fill Missing Values.

```
# Fill missing values (replace missing value in marks)

df["Marks"] = df["Marks"].fillna(df["Marks"].mean())

df
```

	Id	Name	Age	City	Marks
0	101	Rahul	20.0	Surat	85.000000
1	102	Priya	21.0	Navsari	86.142857
2	103	Amit	NaN	Surat	78.000000
3	104	None	20.0	Bharuch	88.000000
4	104	Riya	20.0	Bharuch	88.000000
5	105	kavya	22.0	Navsari	81.000000
6	106	darshana	23.0	Surat	90.000000
7	107	himay	28.0	Ahemedabad	93.000000

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```
# Fill missing Age.
# Replace missing values in the Age column.

df["Age"] = df["Age"].fillna(df["Age"].mean())

df
```

	Id	Name	Age	City	Marks
0	101	Rahul	20.0	Surat	85.000000
1	102	Priya	21.0	Navsari	86.142857
2	103	Amit	22.0	Surat	78.000000
3	104	None	20.0	Bharuch	88.000000
4	104	Riya	20.0	Bharuch	88.000000
5	105	kavya	22.0	Navsari	81.000000
6	106	darshana	23.0	Surat	90.000000
7	107	himay	28.0	Ahemedabad	93.000000

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```
# Fill missing Name.
# Replace missing name with the text "Unknown"

df["Name"] = df["Name"].fillna("Unknown")

df
```

	Id	Name	Age	City	Marks
0	101	Rahul	20.0	Surat	85.000000
1	102	Priya	21.0	Navsari	86.142857
2	103	Amit	22.0	Surat	78.000000
3	104	Unknown	20.0	Bharuch	88.000000
4	104	Riya	20.0	Bharuch	88.000000
5	105	kavya	22.0	Navsari	81.000000
6	106	darshana	23.0	Surat	90.000000
7	107	himay	28.0	Ahemedabad	93.000000

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Step 6: Detect Duplicate Records.

```
# check wheater each row is duplicated.

df.duplicated()
```

	0
0	False
1	False
2	False
3	False
4	False
5	False
6	False
7	False

dtype: bool

```
# Count total no of duplicate record.
df.duplicated().sum()
```

```
np.int64(0)
```

Step 7: Remove Duplicate Records.

```
# Remove duplicate rows from the dataframe.
df = df.drop_duplicates()

df
```

	Id	Name	Age	City	Marks
0	101	Rahul	20.0	Surat	85.000000
1	102	Priya	21.0	Navsari	86.142857
2	103	Amit	22.0	Surat	78.000000
3	104	Unknown	20.0	Bharuch	88.000000
4	104	Riya	20.0	Bharuch	88.000000
5	105	kavya	22.0	Navsari	81.000000
6	106	darshana	23.0	Surat	90.000000
7	107	himay	28.0	Ahemedabad	93.000000

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Step 8: Rename a column.

```
# Rename the column marks to "score".
# Apply the change directly to the original DataFrame using inplace=True.

df.rename(columns ={"Marks":"Score"},inplace=True)

df
```

	Id	Name	Age	City	Score
0	101	Rahul	20.0	Surat	85.000000
1	102	Priya	21.0	Navsari	86.142857
2	103	Amit	22.0	Surat	78.000000
3	104	Unknown	20.0	Bharuch	88.000000
4	104	Riya	20.0	Bharuch	88.000000
5	105	kavya	22.0	Navsari	81.000000
6	106	darshana	23.0	Surat	90.000000
7	107	himay	28.0	Ahemedabad	93.000000

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Step 9: Change Data type.

```
# Change the datatype.
#convert age column float -> integer.

df["Age"] = df["Age"].astype(int)
df.dtypes
```

	0
Id	int64
Name	object
Age	int64
City	object
Score	float64

dtype: object

Step 10: Create a new column.

```
#Create a new column named "final score".

df["Final Score"] = df["Score"]+5

df
```

	Id	Name	Age	City	Score	Final Score
0	101	Rahul	20	Surat	85.000000	90.000000
1	102	Priya	21	Navsari	86.142857	91.142857
2	103	Amit	22	Surat	78.000000	83.000000
3	104	Unknown	20	Bharuch	88.000000	93.000000
4	104	Riya	20	Bharuch	88.000000	93.000000
5	105	kavya	22	Navsari	81.000000	86.000000
6	106	darshana	23	Surat	90.000000	95.000000
7	107	himay	28	Ahemedabad	93.000000	98.000000

Next steps: Generate code with df New interactive sheet

Step 11: Convert text to uppercase.

```
# Convert all student name to uppercase.

df["Name"] = df["Name"].str.upper()

df
```

	Id	Name	Age	City	Score	Final Score
0	101	RAHUL	20	Surat	85.000000	90.000000
1	102	PRIYA	21	Navsari	86.142857	91.142857
2	103	AMIT	22	Surat	78.000000	83.000000
3	104	UNKNOWN	20	Bharuch	88.000000	93.000000
4	104	RIYA	20	Bharuch	88.000000	93.000000
5	105	KAVYA	22	Navsari	81.000000	86.000000
6	106	DARSHANA	23	Surat	90.000000	95.000000
7	107	HIMAY	28	Ahemedabad	93.000000	98.000000

Next steps: Generate code with df New interactive sheet

Step 12: Convert text to lowercase.

```
#Convert city names to lowercase.
df["City"] = df["City"].str.lower()

df
```

	Id	Name	Age	City	Score	Final Score
0	101	RAHUL	20	surat	85.000000	90.000000
1	102	PRIYA	21	navsari	86.142857	91.142857
2	103	AMIT	22	surat	78.000000	83.000000
3	104	UNKNOWN	20	bharuch	88.000000	93.000000
4	104	RIYA	20	bharuch	88.000000	93.000000
5	105	KAVYA	22	navsari	81.000000	86.000000
6	106	DARSHANA	23	surat	90.000000	95.000000
7	107	HIMAY	28	ahemedabad	93.000000	98.000000

Next steps: Generate code with df New interactive sheet

Step 13: Remove Extra spaces.

```
#Remove Extra (white)leading and trailing space from city name.

df["City"] = df["City"].str.strip()

df
```

	Id	Name	Age	City	Score	Final Score
0	101	RAHUL	20	surat	85.000000	90.000000
1	102	PRIYA	21	navsari	86.142857	91.142857
2	103	AMIT	22	surat	78.000000	83.000000
3	104	UNKNOWN	20	bharuch	88.000000	93.000000
4	104	RIYA	20	bharuch	88.000000	93.000000
5	105	KAVYA	22	navsari	81.000000	86.000000
6	106	DARSHANA	23	surat	90.000000	95.000000
7	107	HIMAY	28	ahemedabad	93.000000	98.000000

Next steps: Generate code with df New interactive sheet

Step 14: Sort the Dataset.

```
# Sort data based on final score.
df_sorted = df.sort_values(by="Final Score",ascending=False)
df_sorted
```

	Id	Name	Age	City	Score	Final Score
7	107	HIMAY	28	ahemedabad	93.000000	98.000000
6	106	DARSHANA	23	surat	90.000000	95.000000
3	104	UNKNOWN	20	bharuch	88.000000	93.000000
4	104	RIYA	20	bharuch	88.000000	93.000000
1	102	PRIYA	21	navsari	86.142857	91.142857
0	101	RAHUL	20	surat	85.000000	90.000000
5	105	KAVYA	22	navsari	81.000000	86.000000
2	103	AMIT	22	surat	78.000000	83.000000

Next steps: Generate code with df_sorted New interactive sheet

Step 15: Save the cleaned Dataset.

```
#Save clean dataframe as new CSV file.

df.to_csv("Cleaned_Students.CSV",index=False)
print("Dataset created successfully")
```

```
Dataset created successfully
```